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QatarEnergy

IS SPILL RESPONSE PROCEDURE

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CUSTODIAN DEPARTMENT	DOCUMENT AUTHOR	DOCUMENT OWNER
OIH	Gani Zharikessov Digitally signed by Gani Zharikessov Date: 2024.03.20 08:45:37 +03'00'	Digitally signed by Ibrahim Al-Ahmad Date: 2024.03.21 08:26:26 +03'00'
	OIH/6	OIH

FINAL APPROVAL
Digitally signed by Hamad Al-Baker Date: 2024.03.21 11:33:03 +03'00'
OI

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1.0 OBJECTIVE

To provide a mechanism for the prevention of spills and leaks of both oil and Hazardous and Noxious Substances, as well as detailing the procedures to follow for quick response to any leak or spill, with the purpose of effectively reducing damage to the environment and Asset.

2.0 SCOPE AND APPLICABILITY

This Procedure applies to PS1 Field, employees, contractors, and subcontractors, whether engaged directly by IS or by others, who will perform work or services of any kind at any IS facilities and/or locations.

Contractor drilling rigs, work over, and construction barges working for IS shall in addition, have their own Spill Response Procedure (SPR) or equivalent, which shall incorporate the reporting and control mechanism of this Procedure.

3.0 ROLES AND RESPONSIBILITIES

ON-SCENE COMMANDER

On-Scene Commander (OSC) is the most senior person from IS on site (i.e. OIS, OIM, Supervisor etc.) and:

- Ensure this Procedure is both communicated and implemented on site.
- Regularly inspect the facility on a regular basis to ensure that spill prevention measures are being practiced.
- In case of a spill incident, assess the situation, communicate with support services and authorities, mitigate the source of the incident, and provide resources for spill control and containment. Notify A7S ECC to report the spill.

EMERGENCY RESPONSE TEAM LEADER

- Proceed to incident location and assess the situation by conducting a dynamic risk assessment as instructed by OSC.
- Consult Material Safety Data Sheets for Hazardous and Noxious Substances information, PPE requirements and response strategies. If available, refer to Hazardous and Noxious Substances risk matrix / Tactical Response Plan.
- Be familiar with the Hazardous and Noxious Substances Spill Response PPE, spill kits and other Spill Response equipment, decontamination systems and applications and Hazardous and Noxious Substances Spill Response techniques.
- Brief the response team on the plan, their roles and responsibilities and ensure the Hazardous and Noxious Substances spill response is conducted thoroughly.
- Update the site Hazardous and Noxious Substances risk matrix when new Hazardous and Noxious Substances arrive on site.
- Maintain and correct storage of the PPE and spill response kits.
- Check spill response kits weekly to ensure they are not damaged, used or missing.

QatarEnergy OIL SPILL AND EMERGENCY RESPONSE DIVISION

Responsible for all aspects related to Oil Spill Preparedness, Response and Cooperation for QatarEnergy and the State of Qatar, including:

- Provide the Operational Oil Spill Response and technical input and guidance to IS Incident Management Team (IMT) in formulating the overall Incident Management Strategy upon notification from IS.
- Provide Spill Management Team and technical personnel upon request from IS IMT.
- Deploy oil spill response resources as per pre-established on-water, land and shoreline protection and response strategies in coordination with IS IMT.

DRILLING RIG OFFSHORE INSTALLATION MANAGER (OIM)

- Act as the OSC on the Drilling Rig.
- Be familiar with this Procedure and the contractor Spill Response Plans.
- Regularly inspect the facility to ensure that spill prevention measures are being practiced.

DRILLING SUPERVISOR (DSV)

- Be familiar with this Procedure and the contractor’s Spill Response Plans.
- Inspect the facility on a regular basis to ensure spill prevention measures are being practiced.
- In the event of a spill or release, assess the situation and, if necessary, provide the Drilling contractor with the resources needed for spill mitigation

WELL SERVICES SUPERVISOR

- Ensure that spill prevention measures are implemented in remote jackets.
- At the direction of the OIS or Production Supervisor, initiate any tactical spill response required.

IMT ADVISOR (IS EMERGENCY RESPONSE COORDINATOR (OIH/6)/PLANNER (OIH/61))

- Custodian of this procedure.
- Ensure this procedure is implemented across IS sites.
- In event of Tier 2 / 3 spill, upon notified by A7S ECC, support IMT process as required.

SITE NURSE (MEDIC)

- Provide necessary health screening of ERT Team following spill response activity.
- Provide the medical support for any casualties from the incident.

4.0 PROCEDURAL STEPS

4.1 OIL SPILL RESPONSE

4.1.1 Incident Escalation Levels

As per QatarEnergy Standard for Emergency Preparedness and Response (QP-BCM-STD-011), the incident escalation levels are as follows:

Tier 1: *Event can be dealt with by resources assigned to and deployed by the affected facility/ installation*. Minor operational Spills, either from production operations, wellheads or from sub-sea pipeline transfer operations, which have minor impacts without the potential to escalate and can be managed with local resources. Immediate on-site response capability of Emergency Response Team (ERT), and support vessel personnel may control the spill with resources in place such as absorbent materials for spills on platforms and well heads or mechanical dispersion by using propeller wash if the spill is on water.

Tier 2: *Event can be dealt with by resources deployed by the affected organization, may be supported by internal/external support providers, mutual aid assistance from neighbouring organizations under normal, arrangements*.

Spills that have the potential for moderate to significant impacts and can escalate. Response personnel and equipment can be brought to the scene from the QatarEnergy Offshore Oil Spill Response Base at Halul and from others three QatarEnergy shore-based response locations.

Tier 3: *An event that cannot be controlled with all of the QatarEnergy resources available and requires assistance from Qatar Civil Defence, the wider QatarEnergy organization, or other Government Agencies.*

Spills that have potentially significant impacts at Regional and International Level. The QatarEnergy Tier 3 response organization is activated, including support from Marine Emergency Mutual Aid Center (MEMAC). Response resources and equipment can be brought to the scene from all QatarEnergy stockpiles, and from worldwide spill response cooperative stockpiles such as those available from Oil Spill Response Limited (OSRL).

4.1.2 Alert Procedure, Initial Actions and Notifications

The person at the scene of the operation (Spill Observer) who first identifies an oil release/ spill shall take the following actions:

1. **Alleviate danger**

Take prompt action to alleviate any danger to people, environment or asset. Consider the possible presence of H_2S gas, fire or explosion potential, and pressure hazards. Protect yourself first and inform others. If there is any question about your safety, don breathing apparatus and leave the area by moving across and upwind, if possible.

2. **Notification**

Report the leak or spill immediately by contacting the control room in person by radio or by local call point. Give information about the location, approximate amount of spill, possible cause if it is easy to determine, status of the leak etc. The OSC will be notified of the situation through the chain of command.

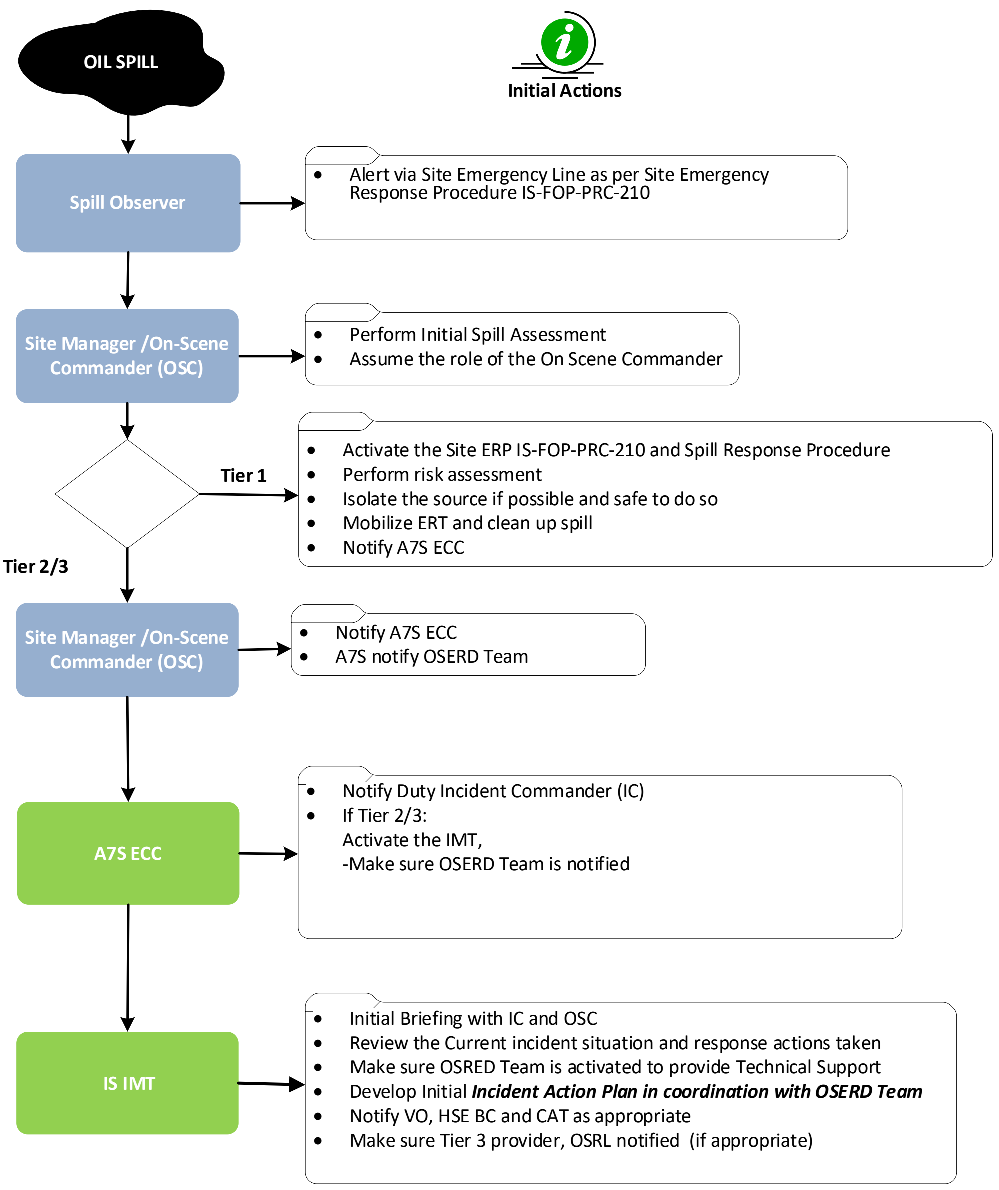
3. Contain Spill or Leak

Take action to safely stop or control the spill or leak source, where feasible and safe to do so.

4. Safety Considerations for Oil Spill Response

- Flammable and Toxic Gas containing H_2S may present at work site.
- Gas monitoring must be conducted at work site / onboard response vessels.
- Personal Gas Monitors must be worn at all time.
- Suitable respiratory protection equipment must be available at all times.
- Site access/egress must be strictly controlled at work site.

FIGURE 1 – OIL SPILL INITIAL NOTIFICATION AND ACTIONS FLOWCHART



4.2 TIER 1 SPILLS

A Tier 1 oil spill response shall be managed by the Site On-Scene Commander (OSC) and an oil spill risk assessment will be conducted at the location. Spill response kits are strategically placed where there is a risk of an oil spill.

In event of the oil spill incident the On-Scene Commander (OSC) shall immediately assess the situation and confirm if spill is Tier 1 and could be handled by Site resources. The following actions shall be taken by the OSC:

- If Tier 1 Spill is confirmed notify A7S ECC.
- Prevent access to the affected area through the normal communication methods.
- Assess the type of substance released (refer to MSDS).
- Mobilize the Emergency Response Team (ERT) in line with Site Emergency Procedure.
- Identify the means of isolation that will safely stop and secure source of leak/ spill, if feasible and activate ESD valves.
- Contain and implement clean up measures using Oil Spill Response Kits.
- Protect facilities by agitating & moving spill away from structure i.e. with firewater monitor and/or fire hoses if appropriate.
- Ensure waste from the spill response is handled in line with QatarEnergy procedures.
- Continue to monitor & evaluate.
- If the incident escalates, notify A7S ECC for additional support.

4.2.1 Tier 2 / 3 Spill

Tier 2 / 3 oil spill response shall be managed by the IS IMT in Doha supported by the OSERD (Oil Spill & Emergency Response Division) Team.

If spill is beyond Tier 1 and / or Site requires additional resources from Onshore then OSC shall notify A7S ECC about the incident and the status including the initial actions. In the event that an incident cannot be effectively controlled or if the circumstances surrounding the incident change, it may be necessary to escalate the response from Tier 1 to Tier 2. This escalation typically occurs when the resources and strategies initially allocated for Tier 1 response are insufficient to address the evolving situation. Tier 2 response involves additional resources, personnel, and strategies to manage the incident effectively and mitigate its impact.

The IS Incident Commander shall request A7S ECC to mobilize Doha based IMT and activate IS Emergency Management Procedure (IS-HSE-PRC-001).

Once OSERD is notified by A7S of a Tier 2/3 oil spill incident, the OSERD (Oil Spill Response Teams – Halul, Mesaieed, Ras Laffan, & Dukhan) shall be activated. Further analysis and evaluation of available information regarding status of spill source control, magnitude, extent, probable impact and response safety for the spill incident shall be conducted. Oil Spill Response Strategies and corresponding operational response shall be established and initiated by OSERD Division based on type of product spilled, volume, safety considerations, location and trajectory and fate modelling, response priorities, (i.e. threat to sensitive resources), weather conditions, available resources, etc.

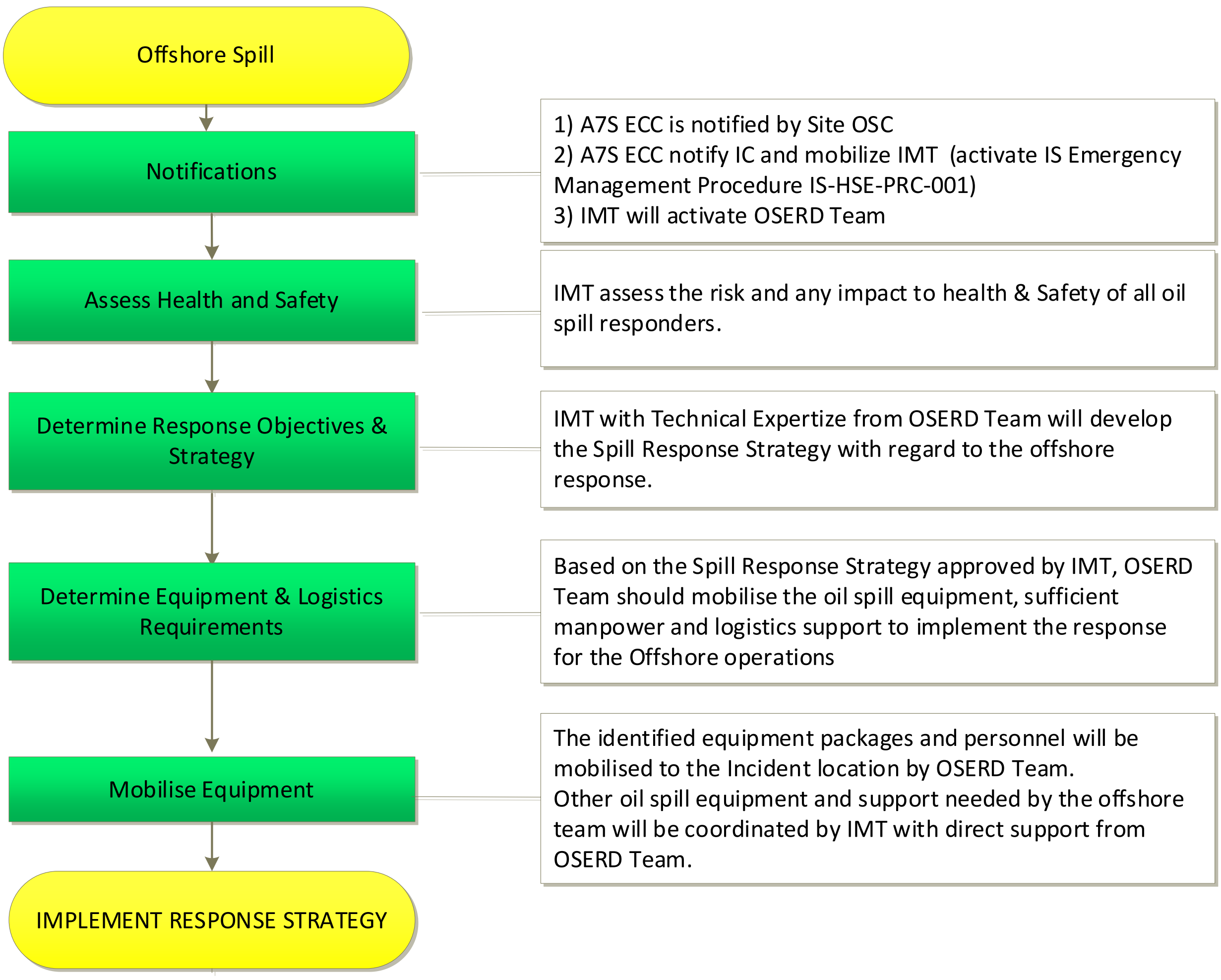
OSERD Team shall initiate response to the incident and establish in conjunction with the IS IMT the response strategy and update on status of response team(s) deployment.

IS IMT shall support with the development of the Incident Action Plan and the overall incident response management in coordination with OSERD team. Initial strategy for any incident occurring at IS offshore facility/location is to launch response from the OSR Base at Halul, supported by other OSERD Response Base locations.

For Tier 3 Spills QatarEnergy may mobilize OSRL by QatarEnergy Nominated Call-Out Authorities from OSERD Team (ORS). Further to the support from OSRL, additional external support can be obtained from Regional Mutual Aid Agreements between the State of Qatar and Marine Emergency Mutual Aid Center (MEMAC). The main purpose of this Regional Organization is to provide support to member States and to facilitate co-operation in order to combat pollution by oil and other harmful substances resulting from marine emergencies. This involves coordinating resources and expertise to contain, mitigate, and clean up spills, minimizing environmental damage. MEMAC also facilitates the sharing of best practices, conducts training and exercises.

The following is the initial response flowchart for Tier 2/3 spills:

FIGURE 2 – OIL SPILL RESPONSE FLOWCHART



<u>Visual calculation of the Spill</u>

A visual estimation can be attained based on the relationship between observed oil colour and its thickness multiply by area. This can be achieved by taking observations directly from the Infield Vessel or a surveillance aircraft. Appendix B provides detailed guidance using the Bonn Agreement Oil Appearance Code (BAOAC).

4.2.2 Activation of QatarEnergy Oil Spill and Emergency Response Team

The following is the extract from the Procedure for Emergency Response to Oil Spills (ORS-MAR-PRC-001). The key features of this procedure are summarized below for reference:

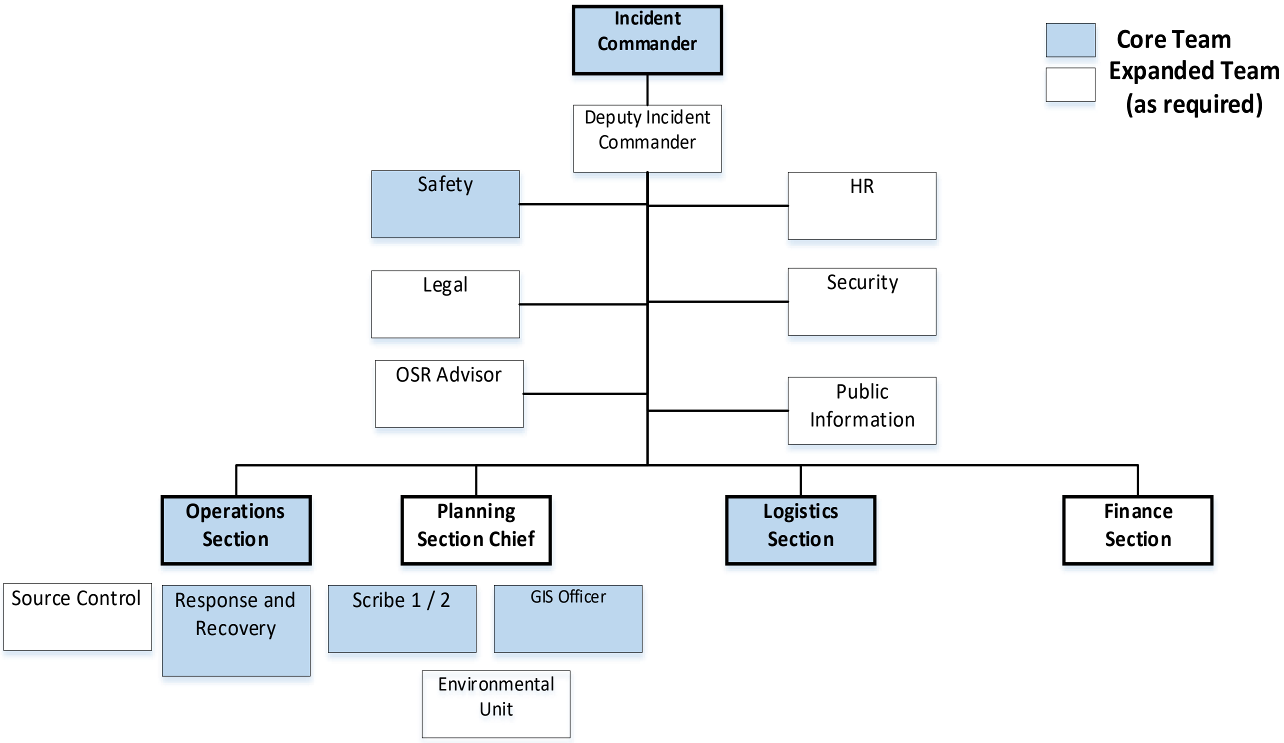
- Verbal Oil Spill notifications shall be provided by the OSC / Incident reporter to A7S ECC on phone No: 4013 3999 (Emergency – when immediate assistance is required); or by phone No: 4013 3888 (for notification purpose – if no assistance required). In addition, verbal notifications shall be followed with a completed Oil Spill Notification Form through email: A7S@qatarenergy.qa or Fax No. 40139155.
- A7S in turn notifies the 'Duty Pollution Officer (DPO)'.
- The DPO is appointed by the QatarEnergy Oil Spill and Emergency Response Division–(OSERD) and is on 24-hours standby /backup duty as per the Duty Roster available on QatarEnergy Intranet.
- The DPO shall establish contact with the OSC / Incident Reporter to request additional information regarding the status of the incident, communicate required information and provide updates to the OSERD Team.
- The OSERD Team initiates and implements oil spill response strategies and actions as per Oil Spill & Emergency Response Division procedures.

4.2.3 Structure of IS IMT for Tier 2 /3 Oil Spills

For Tier 2 and 3 events activate the IS Emergency Management Procedure IS-HSE-PRC-001 and mobilize IS IMT. Technical Support Teams from different QatarEnergy Assets i.e. OSERD, Marine, Security, HR, Legal, Public Relations shall be co-located into OI IMT to provide necessary coordination and support to manage the incident.

Below is a typical IMT structure for Tier 2 /3 oil spill response. Structure of the IMT may be adjusted based on actual incident severity. Who and how many depends on the incident nature.

FIGURE 3 – IMT TYPICAL STRUCTURE FOR TIER 2/3 OIL SPILL INCIDENTS



4.2.4 Oil Spill Response on Drilling, Well Servicing & Workover Barges

Equipment varies depending on the rig or barge being used in the field, but typically the spill response equipment onboard is limited to local use.

Drilling rigs, well servicing and workover barges working for IS shall have their own OSR Plan or equivalent, consistent with the guidance in this Spill Contingency Procedure. In the event of a contained spill on board of the affected rig or barge shall refer to their individual OSR Plan for appropriate spill response.

4.3 HAZARDOUS AND NOXIOUS SUBSTANCES SPILL RESPONSE

In the event of a spill/release, personnel shall firstly ensure their own personal safety and address any potential safety implications to others on or near to the scene.

The key to an effective and safe response is understanding the properties of the Hazardous and Noxious Substances so that the appropriate precautions and actions can be taken. If you are not familiar with the Hazardous and Noxious Substances Hazardous and Noxious Substances and its properties, or **if the substance is unknown** to you then;

- Evacuate the area by moving across and upwind to a place of safety;
- Report the spill immediately by contacting the Control Room and reporting the size, scope, location, type and quantity of the spill/release;
- Take reasonable steps to prevent others from entering the area by isolating the affected area, denying entry through warning and informing others. Consideration should be given to fire or explosion potential, volatile toxic

compounds, and Hazardous and Noxious Substances that pose direct contact hazards.

4.3.1 Initial Response

Notification

After being notified, the OIS/DSV or Facility Supervisor shall determine if additional response action and/or notification is necessary. This shall be based on the estimated size of the spill, the possibility of safely containing the spilled Hazardous and Noxious Substances Hazardous and Noxious Substances, response capabilities, and the spill's potential to negatively impact human life, marine life, equipment or facilities. Priority shall be given to safely stop and secure the source of the Hazardous and Noxious Substances leak.

On Scene Commander shall;

- Mobilize site Emergency Response Team (ERT) as per Site Emergency Response Procedure
- In case of Tier 2 spills,
  - Notify A7S for offshore spills
  - RLIC - IMT for large spills in RLIC facilities (Warehouse)
  - Halul PWHF – notify Halul emergency services
- For Tier 2/3 event Incident Commander to mobilize IMT as per IS Emergency Management Procedure (IS-HSE-PRC-001).

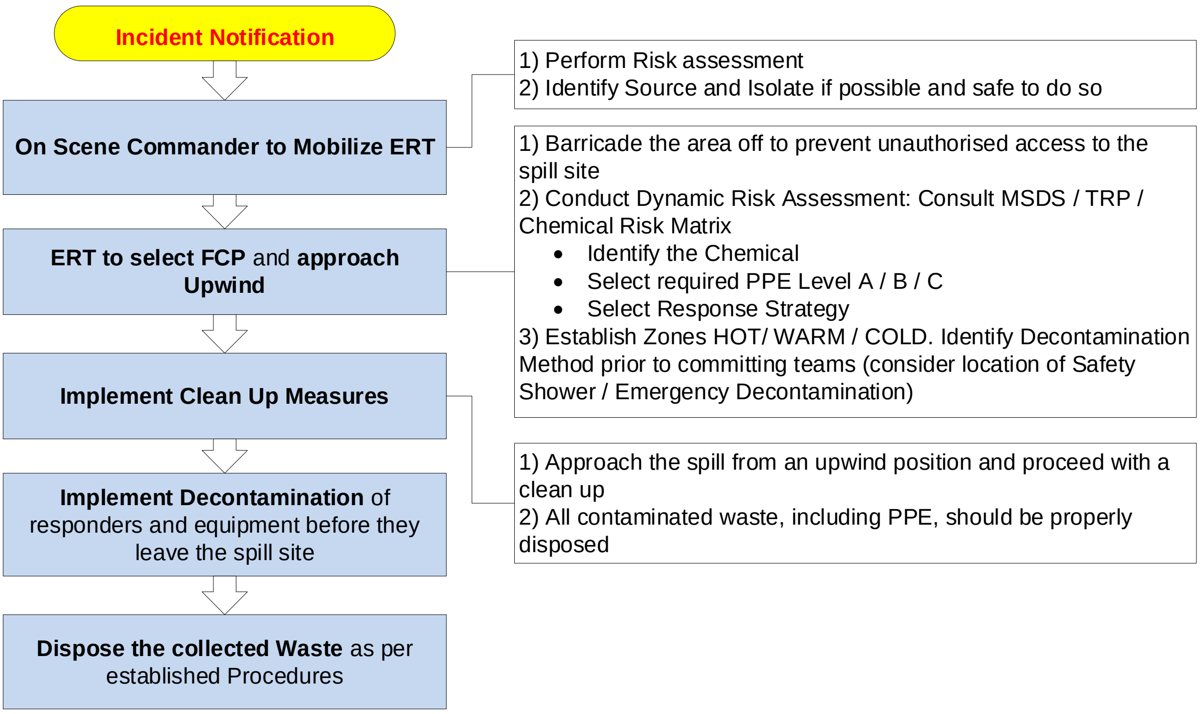
Response Actions

Before responding to a Hazardous and Noxious Substances spill, personnel should have knowledge and understanding of;

1. The potential for harm and routes of exposure;
2. Specific PPE requirements;
3. Decontamination requirements;
4. Product control & mitigation methods;
5. Clean-up and disposal requirements.

Below is a Hazardous and Noxious Substances Spill Response Flowchart with the basic guidance and actions.

FIGURE 4 – CHEMICAL SPILL RESPONSE FLOWCHART



A spill response shall not commence until the Hazardous and Noxious Substances involved has been identified and confirmed. The following are the basic spill response measures:

- If possible and safe to do so, isolate the spill source by remote means.
- Identify spilled Hazardous and Noxious Substances and perform risk assessment (refer to MSDS and/or if available Hazardous and Noxious Substances Risk Matrix / Tactical Response Plan)
- Barricade the incident location and ensure the area is clear of all non-essential personnel
- Select Forward Command Post and establish decontamination method and corridor (Zoning)
- Select and don the correct PPE (A/B/C) according to the information found in the MSDS
- Approach the spill from upwind and contain using absorbent materials (in spill kits)
- Once all of the Hazardous and Noxious Substances is recovered and the site is clear, bag all contaminated materials / PPE into the hazardous waste bags and seal them with tape
- Decontaminate all personnel and equipment before they are removed to the Cold Zone
- Dispose the waste bags in line with the site waste management procedures

4.3.2 Hazardous and Noxious Substances Spill Identification

ERT members shall be familiar with the types, quantity, location and properties of hazardous materials on site. Appropriate signage and identification markings should be present at each location and a Hazardous and Noxious Substances inventory shall be maintained by the site responsible person together with current MSDS for each Hazardous and Noxious Substances or substance.

A **Hazardous and Noxious Substances risk matrix** should be available on Site with all Hazardous and Noxious Substances, their quantities, locations and response actions to be pre-identified in advance of any incident occurring. The Template for Hazardous and Noxious Substances risk matrix can be found at Appendix A. The Hazardous and Noxious Substances risk matrix shall be considered as a live Document and therefore updated on a regular basis by the Site Safety Advisor. Any changes shall be reported to the person responsible for this Document. The Hazardous and Noxious Substances risk matrix shall be included in the site Tactical Response Plans to provide guidance for ERT.

Material Safety Data Sheet

MSDS should be available at site close to the Hazardous and Noxious Substances Storage and Spill Kits. Also, MSDS database is available on HSE K-Drive (K:\Cross-ISND\MSDS\ IS Warehouse).

On receiving notification to respond to a spill/release, responders shall immediately refer to the Hazardous and Noxious Substances Material Safety Data Sheet (MSDS) to be used as the initial method of obtaining information on the hazardous nature of the Hazardous and Noxious Substances, the PPE required and the response strategy to be used. Response depends upon the material physical state (solid, liquid, vapor or gas) and the specific hazards of the Hazardous and Noxious Substances (toxic, corrosive, irritant, carcinogenic etc).

Potential routes of exposure include;

- Inhalation – vapors, gases depending on concentrations;
- Absorption – skin contact hazards;
- Ingestion – orally;
- Injection – skin punctures of pressurized materials or contaminated surfaces.

In general, all MSDS are identical in format, they may however vary slightly dependent on the manufacturer. They all however have the following sections that will be essential in identifying the risks, PPE required, first aid measures, spill measures and disposal measures amongst other items.

Sections of all MSDS – Highlighted the most useful for Hazardous and Noxious Substances spill response.

TABLE 1 – TYPICAL MSDS STRUCTURE.

1.Product Identification	9.Physical and Hazardous and Noxious Substances properties
2.Hazard Identification	10.Stability and reactivity
3.Composition of the Hazardous and Noxious Substances	11.Toxicological information
4.First Aid Measures	12.Ecological information

5.Fire Fighting Measures	13.Disposal considerations
6.Accidental release measures	14.Transport information
7.Handling and storage	15.Regulatory information
8.Exposure control / PPE	16.Other information

Any new Hazardous and Noxious Substances arriving on site shall be reported to site focal point and Site Safety Advisor.

At times there may be Hazardous and Noxious Substances being used onsite that have not been included within this Document, for example during temporary maintenance periods. MSDS shall be provided with these ad-hoc Hazardous and Noxious Substances and consulted prior to commencing a spill response.

4.3.3 PPE Selection

No Hazardous and Noxious Substances garment provides protection from everything. The selection of Hazardous and Noxious Substances protective clothing shall be pre-planned and be dependent upon state and properties of the hazardous materials in use or storage. The level of PPE may also be determined based on a risk assessment of the quantity and nature of spill.

Responders should be familiar with the limitations of each type of Hazardous and Noxious Substances protective clothing and the additional safety precautions required to mitigate the risk of heat exhaustion; reduced communications; restricted vision; change in level; potential time required for decontamination. ERT Team Leader (ERTL) should also ensure that a buddy system is adopted for responders entering the hot zone and that additional responders are available for Rapid Intervention if required.

The PPE level selected for a response to a Hazardous and Noxious Substances spill shall be decided by OSC and the ERTL after consultation of the Safety Data Sheet / Hazardous and Noxious Substances Risk matrix (Appendix A) or Tactical Response Plan (if available).

There are three levels of PPE that need to be considered when dealing with Hazardous and Noxious Substances response:

- **Level A** – provides the highest level of full-body protection to vapors/gases and usually come with integral Hazardous and Noxious Substances resistant gloves and boots (gas-tight-suits). They require breathing apparatus to be worn inside the suit but are proposed for a limited duration use (<20min) due to heat-related physiological effects on the wearers. (Level A PPE is available for the site Emergency Response Team only).
- **Level B** – provides liquid-splash protection but does not protect from gases or vapors. Level B PPE may have integral gloves and/or boots/socks and may be encapsulating or require BA to be worn on the outside. (Level B PPE is available for the site Emergency Response Team only).
- **Level C** – comprises of Hazardous and Noxious Substances resistant clothing, gloves and boots, and disposable respirator / filter. (Level C PPE is available in the Spill Response Kits)

Disposable respirators can be substituted with a positive pressure SCBA set, enhancing the respiratory protection if required.

TABLE 2 –PPE COMPONENTS REQUIRED FOR EACH LEVEL.

Personal Protective Equipment	**A**	**B**	**C**
SCBA			
Disposable Respirator (i.e. N95)			
Totally encapsulated Hazardous and Noxious Substances-protective suit with vapour barrier			
Hooded Hazardous and Noxious Substances resistant suit (one or two piece)			
Coveralls			
Outer Gloves, Hazardous and Noxious Substances resistant			
Inner Gloves, Hazardous and Noxious Substances resistant			
Steel Toe capped boots, Hazardous and Noxious Substances resistant			
Hard hat (worn under suit)			
Hard hat			
Other PPE as required by site risk assessment prior entry			

Requirement
Optional
Inside Spill Kit

PPE Donning

Incorrect use or wearing off PPE can allow Hazardous and Noxious Substances vapours or liquids to come into contact with the user and can cause serious injury. ERTL shall ensure that entry teams are provided assistance to don PPE correctly.

It is essential that the conditions of the working environment are included in the response strategy and that extra precautions are taken to ensure that responders are not overcome by heat stress. When wearing Hazardous and Noxious Substances protective clothing, minimize the working time, provide shelter for all workers, provide reliefs, ensure regular breaks are taken and refreshments are available at all times.

4.3.4 Forward Command Post and Zoning

Emergency Response Team Leaders shall mobilize to a pre-determined Forward Command Post (FCP) that is;

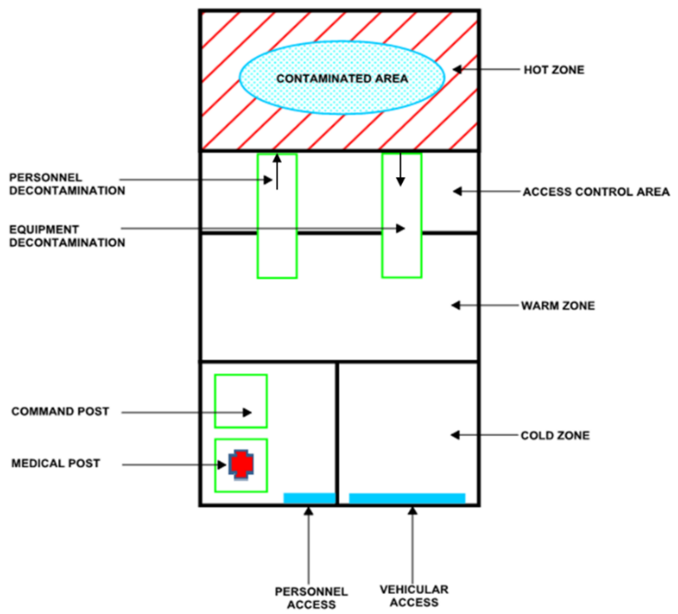
- located upwind from the incident;
- provided with a Hazardous and Noxious Substances PPE cabinet/spill kit;
- near to a safety shower;
- near to an eyewash station.

From the FCP the Team Leader shall conduct a further dynamic risk-assessment of the incident and secure the incident scene by establishing control zones;

- **Hot zone** – Contaminated area where the spill has occurred, and the selected level of PPE must be worn. May include a wider area downwind.
- **Warm zone** – Area between the hot and cold zone where the decontamination zone is established. The decontamination corridor provides the pathway between the contaminated area and the clean area. Personnel assisting in decontamination must wear a suitable level of PPE no more than one level below the hot zone entrants.
- **Cold zone** – Clean area with no further risk of any contact with Hazardous and Noxious Substances. The FCP shall be established in this area.

The diagram below suggests the generic set up for a Hazardous and Noxious Substances spill response, onshore. For offshore, the zoning would depend on the actual layout of the incident location, but the basic principles should be applied. The working area shall be strictly controlled by the ERTL at all times.

FIGURE 5 – ZONING OF THE INCIDENT LOCATION (ONSHORE)



4.3.5 Control and Mitigation Measures

Responders shall not work alone within the hot zone with a minimum of two required for each entry team. ERT Leader shall select the most appropriate and effective method for controlling the spill/release through risk assessment. The chosen solution should minimize the impact on people, assets and the environment:

- **Containment** – stopping the material from leaking from its container by isolating, patching or righting an overturned container.
- **Confinement** – keeping the material within the area of release by damming, diverting, interceptors or confining vapors;
- **No Action** – Create a safe area and allow the product to stabilize whilst protecting any exposures.
- **Absorption** – soaking up the spilled material with pads, socks, etc.
- **Adsorption** – when the Hazardous and Noxious Substances adheres (sticks) to the surface of a material such as sand;
- **Dilution** – addition of water to weaken the strength of the Hazardous and Noxious Substances (caution as this increases the volume of spill)
- **Vapor Dispersion** – using hose streams or monitors to spread vapors and lower concentration;
- **Vapor Suppression** – controlling vapors with water sprays, foam or gelatinizes agents.

4.3.6 Decontamination Requirements

The method required for decontamination of responders should be identified, selected and established **PRIOR** to committing responders into a Hot Zone.

A means for emergency decontamination shall be established as a minimum. For example, a dedicated charged line of hose laid out to the edge of the identified warm zone. The emergency decontamination provision should allow for the rapid removal of contaminants from an ERT member without formally establishing a decontamination corridor.

A decontamination corridor on site may comprise of a nearby safety shower with an adjacent designated undressing area. However, the corridor should be:

- located upwind from the spill;
- between the hot and cold zones;
- remote enough from the cold zone to avoid aerosol contamination from droplets;
- close enough to the hot zone to prevent spreading contaminants further.

The decontamination method shall be determined from the material MSDS and a member of ERT shall be assigned to assist an operator to remove Hazardous and Noxious Substances clothing following decontamination. In general, drenching with low-pressure high-volume water will be a sufficient method, but is essential that this is verified with the MSDS first.

4.3.7 Emergency Decontamination

If personnel working in the area have inadvertently been contaminated, whilst wearing standard PPE, they shall require emergency decontamination.

Other instances where this may be necessary include;

- Damaged or incorrect use of Hazardous and Noxious Substances PPE.
- Further spill, release or rapid escalation of the incident requiring teams to immediately vacate the hot zone.

If this does occur, and taking into consideration your safety and the safety of others.

- Remove contaminated clothing from affected person.
- If possible, move the affected person to the nearest safety shower for decontamination (consult MSDS to confirm that water may be used for decontamination).
- If no safety shower is nearby or the affected person is incapacitated in any way, a fire hose on reduced pressure may be used for decontamination.
- Seek medical attention immediately and provide the medical staff with the MSDS.

4.3.8 Waste Disposal

Waste from clean-up materials and/or recovered Hazardous and Noxious Substances, should be collected, contained, and labelled appropriately as per the Hazardous and Noxious Substances MSDS, and sent to Ras Laffan IS Warehouse (Laydown area), for proper treatment and further disposal actions as per Waste Management Procedure (IS-HSE-PRC-012).

4.4 TRAINING AND DRILLS

To ensure that all site personnel are familiar with the oil spill kits and the actions required to be taken, training sessions shall be conducted for site personnel who work in the areas at risk. The training shall be conducted by Site Safety representative on site (ERT).

Spill training should include the following:

- Spill identification
- Use of Material safety data sheets / Hazardous and Noxious Substances Risk Matrix / Tactical Response Plans
- Awareness of the spill kits and their use
- PPE requirements
- Product control and mitigation measures
- Decontamination methods
- Waste disposal
- Practical demonstration of a site setup (zoning) and using the spill kit contents

The Site ERT shall conduct regular Site drills (Tier 1) simulating a Hazardous and Noxious Substances response incident, with different Hazardous and Noxious Substances and utilizing different types of equipment on Site and MSDS and / or (if available) the Hazardous and Noxious Substances Risk Matrix / Tactical Response Plans. The spill incident response shall be included in the Site Drill Schedule.

Tier 2 and 3 Drills shall be conducted by IMT in coordination with Site and OSERD Team.

4.5 REPORTING OF SPILLS

Spills shall be reported as per the following:

- QatarEnergy Procedure for HSE Incident Reporting, Investigation & Learning (QP-HSE-PRC-022)
- QatarEnergy Standard for HSE Incident Reporting, Investigation & Learning QP-HSE-STD-021
- Reporting of the Spills to The Ministry of Environment and Climate Change (MoECC), as per the CTO requirements.

5.0 REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal /External	Doc Referencing
IS Emergency Management Procedure	IS-HSE-PRC-001	Internal	Upward
Waste Management Procedure	IS-HSE-PRC-012	Internal	Sideways
PS-1 Emergency Response Procedure	IS-FOP-PRC-210	Internal	Sideways
Infield Pipeline Emergency Response Procedure	IS-FOP-PRC-250	Internal	Sideways
Halul Produced Water Handling Facility Emergency Procedure	IS-FOP-PRC-410	Internal	Sideways
QatarEnergy Procedure for HSE Incident Reporting, Investigation & Learning	QP-HSE-PRC-022	Internal	Sideways
QatarEnergy Standard for HSE Incident Reporting, Investigation & Learning	QP-HSE-STD-021	Internal	Sideways
National Contingency Plan for the Response to Spills of Oil and Hazardous & Noxious Substances	N.A.	External	N.A.
QatarEnergy Standard for Emergency Preparedness and Response	QP-BCM-STD-011	Internal	Upwards
Procedure for Emergency Response to Oil Spills	ORS-MAR-PRC-001	Internal	Sideways
Oil Spill Notification Form	ORS-MAR-PRC-001-F	Internal	Sideways
Halul Oil Spill Response Plan	QPR-OMO-80	Internal	Sideways
Ras Laffan Oil Spill Response Plan	QPR-OMO-79	Internal	Sideways

APPENDIX

APPENDIX A – REVISION HISTORY LOG

Revision Number: 03 **Document Revision Date: 26/12/2023**

Item Revised	Revision Description	Page No.
Miscellaneous	• Bi-annual review of document • Alignment of terminology and reference	All
Remarks:		

APPENDIX B – LIST OF ABBREVIATIONS

Abbreviation	Description
A7S	QatarEnergy Communication Centre Located at HQ, T7.L3, West Bay, Doha
CTO	Consent to Operate (Issued by MoECC)
DPO	Duty Pollution Officer (24 hours standby/back up duty)
ECC	Emergency Communication Centre
DSV	Drilling Supervisor
ESD	Emergency Shutdown
IMT	Incident Management Team
F&G	Fire and Gas
IC	Incident Commander
MoECC	Ministry of Environment and Climate Change
MSDS	Material Safety Data Sheet
OSC	On-Scene Commander
OIS	Offshore Installation Superintendent
PPE	Personal Protective Equipment
IS	IDD El Shargi Asset of QatarEnergy

Abbreviation	Description
QatarEnergy-OSERD	QatarEnergy Oil Spill and Emergency Response Division
RLIC	Ras Laffan Industrial City
PWHF	Produced Water Handling Facility at Halul Island
ROPME	Regional Organization for Protection of The Marine Environment
SPR	Spill Response Procedure
Shall	Mandatory action

APPENDIX C – HAZARDOUS AND NOXIOUS SUBSTANCES RISK MATRIX (SAMPLE)

Hazardous and Noxious Substances Name	Hazardous Classification	PPE Class SMALL SPILL	PPE Class LARGE SPILL	Use spark proof and explosion proof tools	Do not allow to dry out, flush thoroughly with water	Contain	Neutralize with appropriate acid/base (refer to SDS)	Absorb with an inert dry material	Flammable vapors may develop	Evolving gases spread at ground level	Suitable extinguishing media	Location	Volume
Sodium Metabisulfite food grade (E223)	Harmful to Health	C	C								Dry Hazardous and Noxious Substances, CO2, water spray (fog), foam		
Hypochlorite Solution	Non-Flammable	C	B										
Sodium Hypo Chlorite (Liquid Bleach)		C	C										

APPENDIX D - HYDROCARBON SPILL CALCULATION

Spill Size Estimation

Calculate Spill Coverage

If the source / quantity is unknown, then a visual estimation can be attained based on the relationship between observed oil colour and its thickness and area using the Bonn Agreement Oil Appearance Code (BAOAC). This can be achieved by taking

Average Width (km)		km	Average Length in (km)		km

STEP 1. Total area: Estimate total size of the area as a square or rectangle (in km^2).

STEP 2. Oil Spill Area: Assess the area affected by the slick in km^2 calculated as a % of the total area (i.e. 90% of 20 km^2 = 18 km^2).

STEP 3. Calculate area by colour: Estimate the area covered by each colour of oil as a % of area affected in km^2 (e.g. 60% silvery, 40% metallic = 10.8 km^2 and 7.2 km^2 respectively)

STEP 4. Calculate quantity by colour: Multiply the area covered by each colour (Min and Max) by the appropriate quantity of oil in the table (e.g. 10.8 km^2 x 0.04 and 0.3 for silvery and 7.2 km^2 x 5 and 50 for metallic).

STEP 5. Total quantity: Add all the quantity by colour figures to get total quantity of oil/m^3.

STEP 6. Conversion: If necessary, you can covert m^3 to tonnes by multiplying total quantity in m^3 by the Specific Gravity of the spilt oil.

Step 1	Total Area (Width x Length) km^2	20km2	km^2
Step 2	Oil Spill Area (Estimated) km^2	18km2	km^2

Colour	Code	Minimum (m^3 / km^2)	Maximum (m^3 / km^2)	(Step 3) % of Area	(Step 3) Area
Oil Sheen Silvery	1	0.04	0.3	60%	10.8 km2
Oil Sheen Rainbow	2	0.3	5.0	n/a	n/a
Oil Sheen Metallic	3	5.0	50	40%	7.2 km2
Discontinuous True	4	50	200	n/a	n/a
Continuous True	5	200	>200	n/a	n/a

Calculation for Area Covered: km^2 = Area / 100 x % of Area Covered.

Colour	(Step 3)	(Step 4)	(Step 4)

Oil Sheen Silvery	10.8 km2	0,432	3,24
Oil Sheen Rainbow	n/a	n/a	n/a
Oil Sheen Metallic	7.2 km2	36	360
Discontinuous True	n/a	n/a	n/a
Continuous True	n/a	n/a	n/a
Step 5	Total Volume (m^3)	~400m3	
Step 6	Total Volume in Tonnes (m^3 x SG)		

Note:
This calculation method is used for the mitigation purpose, it's not used to determine the amount of oil spilled from the source. When oil is spilled into the waters, it will experience weathering processes (dispersion, evaporation, emulsification, etc.) so that the appearance of the colour and extent of the spilled oil on the water surface will change over time. Therefore, the calculation using this method is the estimated volume of the spill at the time the observation was made. The calculation will show different results if the observation is carried out at a different time.

BONN AGREEMENT OIL APPEARANCE COLOUR CODE

Code	**Example**	**Description**
Code 1 Oil Sheen Silvery (< 0.3 µm)		The very thin films of oil reflect the incoming light better than the surrounding water and will be a silvery or grey sheen. Above a certain height or angle of view the observed film may seem to disappear.
Code 2 Oil Sheen Rainbow (0.3 µm – 5.0 µm)		Rainbow oil appearance is caused by an optical effect and is independent of oil type. Depending on angle of view and layer thickness, the distinctive colours will be diffuse or be very bright. Bad light conditions may cause the colours to appear duller. A level layer of oil in the rainbow region will show different colours through the slick because of the change in angle of view. Therefore, if rainbow is present, a range of colours will be visible.
Code 3 Oil Sheen Metallic (5.0 µm – 50 µm)		Although a range of colours can be observed (e.g. blue, purple, red and greenish) the colours will not be like 'rainbow'. Metallic will appear as a quite homogeneous colour that can be blue, brown, purple or another colour. The 'metallic' appearance is the common factor and has been identified as a mirror effect, dependent on light and sky conditions. For example, blue, can be observed in blue-sky conditions.
Code 4 Discontinuous True Colours (50 µm – 200 µm)		For oil slicks, thicker than 50 µm the true colour will gradually dominate the colour that is observed. Brown oils will appear brown, black oils will appear black. The broken nature of the colour, due to thinner areas within the slick, is described as discontinuous. Discontinuous' should not be mistaken for 'coverage'. Discontinuous implies true colour variations and not non-polluted areas.
Code 5 True Colours (>200 µm)		The true colour of the specific oil is the dominant effect in this category. A more homogenous colour can be observed with no discontinuity as described in Code 4. This category is strongly oil type dependent, and colours may be more diffuse in overcast conditions.

APPENDIX E – OIL SPILL NOTIFICATION FORM (ORS-MAR-PRC-001-F)

QatarEnergy	
OIL SPILL NOTIFICATION FORM	تقرير عن حادث تسرب نفطي
To: Alfa 7 Sierra Fax: 40139155 Tel: 40133999 (Emergency) **Tel:4013388** (Notification for Alert only) E-mail: a7s@qatarenergy.qa Marine Channel 16 - SSB 2370 KHz – WX-1 Ref.: / / ORS	الى: Alfa 7 Sierra فاكس : 40139155 تلفون : 40133999ا بريد الكتروني : a7s@qatarenergy.qa قناة النداء البحرية : 16 او SB 2370 KHz او WX-1 الرقم : / / ORS
Date: Time of Incident:	التاريخ: وقت الحادث:
Location of the Spill:	موقع التسرب:
Latitude: N	خط العرض: شمال
Longitude: E	خط الطول: شرق
With Reference to Installation or a Landmark (e.g. "100 m to NW" of):	الموقع نسبة الى منشأة نفطية او علامة أرضية (مثلا 100 متر شمال غرب)
Description of Oil Spill:	تفاصيل التسرب النفطي:
Length: M	طول بقعة الزيت: متر
Width: M	عرض بقعة الزيت: متر
Estimated Quantity: M^3	كمية الزيت المتسرب: متر مكعب
Source of Spill (If known):	مصدر التسرب (ان كان معروفا):

Did the leak stop:	هل توقف التسرب:
Cause of Oil Spill (If Known):	سبب التسرب (ان كان معروفا):
Type of Oil Spilled: (Crude, Diesel, Hydraulic Oil, etc.)	نوع الزيت المتسرب: (زيت خام ، ديزل ، زيت هيدروليك ، الخ)
Appearance of Oil Spilled: (Liquid, Floating Solid, Sludge, Tarry lumps)	مظهر الزيت المتسرب: (سائل ، كتل عائمة ، رواسب زيتية ، كتل من القطران)
Color of the Oil Slick: (Sheen, Rainbow, Silver, Black, Orange)	لون بقعة الزيت: (لامع ، قزحي ، اسود ، رمادي ، برتقالي)
Weather Conditions:	أحوال الطقس والبحر:
Wind Speed: Knots	سرعة الرياح: عقدة
Wind Direction: (e.g. N – W)	اتجاه الرياح: (مثلا: شمال- غرب)
Sea State: (Calm, Moderate, High)	حالة البحر: (هادئ, متوسط, عالي)
Air Temperature: °C	درجة حرارة الهواء: °م
Sea Water Temperature: °C	درجة حرارة ماء البحر: °م

Remarks and Additional Information:	ملاحظات ومعلومات إضافية:
Reporter:	قام بالتبليغ:
Name:	الاسم:
Position:	الوظيفة:
Tel:	تليفون:
Mobile:	جوال:
Fax:	فاكس: